25900463 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCICH200 - FUNCTIONAL OPERATIONS IN CHEMICAL INDUSTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. The most common type of tray used in distillation columns is [E] / [CO1]

   a). Valve tray b). Bubble-cap tray c). Sieve tray d). All of the above

2. The mixing mechanism in a tumbler mixer is primarily: [An] / [CO3]

   a). Convective mixing due to tumbling motion b). Shear due to blades

   c). Turbulent eddies d). Sedimentation

3. Bernoulli's principle is based on the conservation of [K] / [CO1]

   a). Mass b). Energy

   c). Momentum transfer d). Heat

4. Which method is also called flame fusion? [K] / [CO2]

   a). Zone refining b). Hydrothermal

c). Flame fusion d). Czochralski

5. Which of the following is NOT true for batch distillation? [E] / [CO1]

a). Suitable for small-scale separation

b). Flexible to handle multiple feed mixtures

c). Requires long downtime between batches

d). Produces constant composition distillate

6. Nutsche filter is a: [An] / [CO1]

a). Continuous filtration technique

b). Batch filtration technique using vacuum or pressure

c). Filtration technique based only on gravity

d). Centrifugal separation method

7. In an MSMPR crystallizer, what does MSMPR stand for? [K] / [CO2]

a). Mixed Suspension Mixed Product Removal

b). Multiple Stage Mixed Phase Reactor

c). Mixed Solvent Mixed Product Removal

d). Mixed Suspension Multiple Product Reactor

8. The function of the wiper blades in a wiped film evaporator is to: [U] / [CO1]

a). Increase liquid residence time b). Spread liquid into a thin turbulent film

c). Maintain constant vapor pressure d). Prevent liquid from boiling

9. The process of transferring mass from one phase to another is called [U] / [CO1]

a). Heat transfer b). Mass transfer c). Momentum transfer d). Adsorption

10. Which of the following is NOT an advantage of forced circulation evaporators? [E] / [CO1]

a). High heat transfer rates b). Suitability for viscous liquids

c). Short residence time d). Reduced fouling

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **3** marks

11. Why is steam distillation used for separating essential oils? [U] / [CO1]

12. Write a note on the solvents used for liquid-liquid extractions. [K] / [CO3]

13. Explain how filter aids increase the rate of filtration. [E] / [CO1]

14. Mention two applications of rotary dryer. [E] / [CO1]

15. Define mixing. Why is it important in chemical and pharmaceutical industries? [An] / [CO3]

16. Explain the principle of the Bridgman technique in crystal growth. [U] / [CO2]

[3x4 = 12]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **7** marks

17. Explain the principle of crystallization in detail. Discuss how solubility, temperature, and pressure affect the process [U] / [CO2]

18. Give a detailed account of the uses of extraction in qualitative and quantitative analysis. [U] / [CO3]

19. During Supercritical Fluid Extraction of hops for beer flavoring, researchers use ethanol as a co-solvent along with CO_2. a)What role does the co-solvent play? b)Why might CO_2 alone be insufficient? [U] / [CO3]

20. Describe in detail the stages of crystallization: supersaturation, nucleation, and crystal growth, with suitable diagrams. [U] / [CO2]

21. Discuss the importance of reflux ratio in rectification. How does it affect the efficiency, separation quality, and energy consumption of the distillation process? [An] / [CO1]

22. Write an essay on how simple distillation illustrates the principle of volatility differences. [U] / [CO1]

[7x4 = 28]